

The empirical recurrence for OEIS A254094, recovered from its tail

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Abstract

OEIS A254094 records “Empirical recurrence of order 64” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 195$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 7$, so the recurrence is proved for every $n \geq 71$. Its last coefficient is $c_{64} = -30983446494107742247059456 \neq 0$, so no further tail satisfies anything shorter; any order-64 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A254094 is “Number of $(3+2)X(n+2)$ 0..1 arrays with every $3X3$ subblock diagonal maximum minus antidiagonal maximum nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

22728, 438463, 9231485, 194360930, 4070778350, 85504231094, 1795818466078, 37718365402050, ...

The entry states:

Empirical recurrence of order 64 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 14:28:02, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 195$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 195$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 64: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 7$ is the first whose minimal order is exactly the stated 64.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 390$ exact terms from $n = 7$ on, it returns order 64 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{64} c_i a(n-i),$$

c_1	25	c_{17}	38409784414208	c_{33}	3091660139153514823680	c_{49}	−535552979
c_2	0	c_{18}	−175680247201792	c_{34}	524883704423066370048	c_{50}	1810728982
c_3	−2072	c_{19}	1175667844079616	c_{35}	21505474271179689689088	c_{51}	2698529016
c_4	10076	c_{20}	−7150169375309824	c_{36}	20648185127010016690176	c_{52}	1203658082
c_5	−92076	c_{21}	6148306515984384	c_{37}	−21355193644268239454208	c_{53}	2663271259
c_6	297584	c_{22}	−46225495523328000	c_{38}	21585587213955510566912	c_{54}	7716986084
c_7	1294896	c_{23}	3960864304267264	c_{39}	−465858365154784903692288	c_{55}	−2685421876
c_8	−25250672	c_{24}	327018935554146304	c_{40}	−180443440645763772710912	c_{56}	−790975984
c_9	559638960	c_{25}	−57430089838624768	c_{41}	−518232975332577665613824	c_{57}	−237273044
c_{10}	−1948627904	c_{26}	5694599961480527872	c_{42}	129186179238825883074560	c_{58}	3481267235
c_{11}	1157672512	c_{27}	−3986594667836735488	c_{43}	4345309952060278397992960	c_{59}	3872796334
c_{12}	5742555904	c_{28}	14666428197185257472	c_{44}	3738239325863438169669632	c_{60}	1357677412
c_{13}	−505470546432	c_{29}	−62706531971450798080	c_{45}	590831867946561423015936	c_{61}	2915456249
c_{14}	1957020427776	c_{30}	−180204855252433764352	c_{46}	4935891566111400608661504	c_{62}	
c_{15}	−4871652137472	c_{31}	−201530757133302235136	c_{47}	−64735197556969974936895488	c_{63}	−877864317
c_{16}	25838688849920	c_{32}	−1186823818542929936384	c_{48}	−8848477839794210758197248	c_{64}	−309834464

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 71$, and it is the recurrence OEIS A254094 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 7$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{64} - c_1 x^{64-1} - \dots - c_{64}$. Its constant term is $-c_{64} = 30983446494107742247059456$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no

component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 64 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 64, so $\deg q = 64$, and it is monic. It holds from some index t on. From $\max(t, 7)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 64, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 64, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is $-30983446494107742247059456$.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-64 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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